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Project Report

On

“Big Data Processing and Analysis of US Accidents Dataset”

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“Big Data Processing and Analysis of US Accidents Dataset”

Abstract

This project presents a practical implementation of Big Data analytics using the US Accidents dataset and the Hadoop ecosystem. The dataset contains a large number of real-world accident records from different states of the United States. It includes important information such as accident severity, time, geographic location, distance, weather conditions, visibility, and traffic-related factors. The large size and variety of the dataset make it suitable for demonstrating Big Data storage, processing, and analytical techniques.

The project uses Hadoop Distributed File System (HDFS) to store and manage the accident dataset. After uploading the dataset to HDFS, Java-based MapReduce is used to perform five analytical tasks. These tasks analyze accidents based on state, severity, city, weather condition, and traffic signals. The MapReduce programs demonstrate how Mapper and Reducer classes can process and aggregate large amounts of data efficiently. In addition, Apache Pig is used to perform ten analytical tasks on the same dataset. These tasks include filtering severe accidents, state-wise and severity-wise counting, weather and city analysis, average severity calculation, day and night comparison, average distance analysis, timezone analysis, and weather-severity analysis. These operations demonstrate important Pig concepts such as filtering, grouping, counting, averaging, and data aggregation.

Overall, the project demonstrates how HDFS, MapReduce, and Apache Pig can be combined to build a complete Big Data processing workflow. The analyses transform raw accident records into meaningful statistical information about accident frequency, severity, location, weather, and road-related factors. This project therefore provides a practical understanding of using Hadoop technologies for large-scale real-world data analysis.

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1. Introduction

The rapid growth of vehicles, GPS, weather monitoring, and transportation systems generates large amounts of accident data. Analyzing this data can help identify patterns related to accident frequency, severity, location, weather, and road conditions.

This project analyzes the US Accidents dataset using the Hadoop ecosystem. The project uses HDFS for distributed data storage, MapReduce for processing and aggregation, and Apache Pig for high-level data analysis.

Five MapReduce jobs are performed to analyze accidents by state, severity, city, weather condition, and traffic signal. In addition, ten Pig tasks are used for filtering, grouping, counting, averaging, and analyzing different accident characteristics.

Overall, the project demonstrates how Big Data technologies can be used to process a large real-world dataset and extract useful information about road accidents.

2. Problem Statement

The US Accidents dataset contains a very large number of accident records and many descriptive attributes. Performing repeated calculations such as counts by state, severity, city, weather condition, and traffic signal using conventional manual methods is inefficient and difficult to maintain. As the dataset grows, the time and memory required for single-machine processing can also increase significantly.

The problem addressed by this project is therefore how to store and process a large accident dataset efficiently while supporting multiple analytical requirements. A scalable environment is required where the raw dataset can be stored reliably and where independent analytical operations can be executed without manually processing every record. Hadoop provides such an environment through HDFS for storage and MapReduce/Pig for distributed processing.

The project specifically aims to solve this problem by placing the dataset into HDFS and applying five MapReduce jobs and ten Pig analyses. The outputs provide summaries that are easier to interpret than the raw CSV and demonstrate the practical use of distributed Big Data processing.

3. US Accidents Dataset Description

The US Accidents (2016–2023) dataset is a large-scale traffic accident dataset covering 47 states of the United States. It contains accident-related information such as severity, time, location, distance, weather conditions, and road features. The dataset was collected from February 2016 to March 2023 and contains approximately 7.7 million accident records.

Dataset Source: Kaggle

Dataset Link: <https://www.kaggle.com/datasets/sobhanmoosavi/us-accidents>

The CSV format makes it suitable for loading into Hadoop and processing with both MapReduce and Apache Pig.

Field	Description
ID	Unique identifier of an accident record.
Source	Source from which the accident record was obtained.
Severity	Accident severity category.
Start_Time / End_Time	Beginning and ending timestamps associated with the accident.
Start_Lat / Start_Lng	Latitude and longitude of the accident start location.
End_Lat / End_Lng	Latitude and longitude of the accident end location.
Distance	Distance associated with the accident record.
Description	Textual description of the accident.
Street / City / County / State	Geographic location information.
Weather_Condition	Recorded weather condition.
Temperature / Humidity / Pressure / Visibility	Environmental measurements.
Wind_Direction / Wind_Speed / Precipitation	Wind and precipitation information.
Amenity, Bump, Crossing, Give_Way, Junction, No_Exit	Road/environment feature indicators.
Railway, Roundabout, Station, Stop, Traffic_Calming, Traffic_Signal, Turning_Loop	Additional road and traffic feature indicators.
Sunrise_Sunset / Civil_Twilight / Nautical_Twilight / Astronomical_Twilight	Time-of-day and twilight indicators.

The project uses the complete 47-field schema in Pig, with appropriate numeric and character data types. The schema is important because it determines which fields can be grouped, counted, filtered, or averaged during analysis.

4. Project Objectives

- Store the large US Accidents CSV dataset in HDFS and verify that it is accessible.
- Demonstrate Java-based Hadoop MapReduce processing through five independent analytical jobs.
- Use Apache Pig to perform ten higher-level data-flow analyses on the same HDFS dataset.
- Generate separate, verifiable HDFS outputs for every analysis.
- Compare the suitability of low-level MapReduce logic with concise Pig Latin operations.
- Extract meaningful patterns related to accident location, severity, weather, distance, time of day, and road features.
- Document a complete and reproducible Big Data workflow suitable for laboratory evaluation.

5. Tools and Technologies

- **Hadoop 3.2.4** — distributed storage and processing framework.
- **Hadoop Distributed File System (HDFS)** — storage layer for the dataset and outputs.
- **Java/JDK** — implementation language for MapReduce programs.
- **Apache Pig 0.17.0** — high-level data-flow analysis using Pig Latin.
- **Windows Command Prompt** — project execution and administration environment.
- **CSV** — input data format.

The environment was configured so that Hadoop commands, Java compilation, HDFS commands, MapReduce jobs, and Pig scripts could be executed from Command Prompt. During execution, the Windows Hadoop environment required the session variable `HADOOP_USER_NAME=TechSource` to avoid a username/path-space issue in YARN container execution.

6. Preparing the Hadoop Environment

Before processing the dataset, the Hadoop environment was checked and the required services were started. Java was required for Hadoop and MapReduce compilation, Hadoop was required for HDFS and YARN, and Apache Pig was required for the ten Pig analyses. Version checks were performed from Command Prompt. Hadoop was confirmed as version 3.2.4 and Pig as version 0.17.0. The main local project directory was organized as `C:\BigDataProject` with source files under `C:\BigDataProject\src`, compiled classes under `C:\BigDataProject\classes`, and dataset files under `C:\BigDataProject\dataset`.

6.1 Implementing HDFS for US Accidents Dataset

Start Hadoop

Hadoop services were started so that HDFS and YARN were available. The running Java processes were checked with `jps`. The expected services include NameNode, DataNode, ResourceManager, and NodeManager.

```
C:\hadoop\sbin>jps
23780 SecondaryNameNode
17480 Jps
20168 NameNode
21320 NodeManager
6024 DataNode
1244 ResourceManager
```

6.2 Create Project Folder

The local project was maintained under C:\BigDataProject. Separate folders were used for MapReduce, Pig scripts, dataset files, and output-related files to keep the workflow organized.

```
C:\hadoop\sbin>cd /d C:\BigDataProject
C:\BigDataProject>tree /F
Folder PATH listing for volume Windows10
Volume serial number is 963B-8F97
C:..
  .gitignore
  readme.html
  --
  classes
    CityCount$CityMapper.class
    CityCount$CityReducer.class
    CityCount.class
    SeverityCount$SeverityMapper.class
    SeverityCount$SeverityReducer.class
    SeverityCount.class
    StateCount$StateMapper.class
    StateCount$StateReducer.class
    StateCount.class
    TrafficSignalCount$TrafficSignalMapper.class
    TrafficSignalCount$TrafficSignalReducer.class
    TrafficSignalCount.class
    WeatherCount$WeatherMapper.class
    WeatherCount$WeatherReducer.class
    WeatherCount.class
  --
  dataset
    readme.html
```

6.3 Inspect the CSV

The US Accidents CSV files were inspected before HDFS upload. The header was checked to confirm the expected fields and comma-separated structure. Inspecting the raw input before processing is important because malformed headers, missing fields, or damaged files can cause downstream MapReduce and Pig jobs to fail.

```
C:\BigDataProject\dataset>dir
Volume in drive C is Windows10
Volume Serial Number is 963B-8F97

Directory of C:\BigDataProject\dataset

09/06/2026  11:19 PM    <DIR>          .
09/06/2026  10:22 AM    <DIR>          ..
09/05/2026  11:40 PM                12,152  readme.html
09/06/2026  11:18 PM           566,994,377  US_Accidents_1.5M.csv
09/06/2026  11:09 PM       3,058,183,727  US_Accidents_March23.csv
               3 File(s)  3,625,190,256 bytes
               2 Dir(s)  42,057,560,064 bytes free
```

6.4 Create HDFS Directories

The main HDFS workspace was created under /usaccidents. The input directory was created at /usaccidents/input and separate output directories were used for MapReduce and Pig results.

Example commands:

```
set HADOOP_USER_NAME=TechSource
```

```
hdfs dfs -mkdir -p /usaccidents/input
```

```
hdfs dfs -mkdir -p /usaccidents/output_state
```

```

C:\BigDataProject\dataset>hdfs dfs -mkdir -p /usaccidents/input
C:\BigDataProject\dataset>hdfs dfs -ls /usaccidents
Found 1 items
drwxr-xr-x - Tech Source supergroup 0 2026-09-06 23:27 /usaccidents/input
C:\BigDataProject\dataset>hdfs dfs -put US_Accidents_1.5M.csv /usaccidents/input/
C:\BigDataProject\dataset>hdfs dfs -ls -h /usaccidents/input
Found 1 items
-rw-r--r-- 1 Tech Source supergroup 540.7 M 2026-09-06 23:28 /usaccidents/input/US_Accidents_1.5M.csv
C:\BigDataProject\dataset>hdfs dfs -ls -h /usaccidents/input
Found 1 items
-rw-r--r-- 1 Tech Source supergroup 540.7 M 2026-09-06 23:28 /usaccidents/input/US_Accidents_1.5M.csv
C:\BigDataProject\dataset>powershell -NoProfile -Command "Get-Content 'US_Accidents_1.5M.csv' -TotalCount 2"
ID,Source,Severity,Start_Time,End_Time,Start_Lat,Start_Lng,End_Lat,End_Lng,Distance(mi),Description,Street,City,County,S
tate,Zipcode,Country,Timezone,Airport_Code,Weather_Timestamp,Temperature(F),Wind_Chill(F),Humidity(%),Pressure(in),Visib
ility(mi),Wind_Direction,Wind_Speed(mph),Precipitation(in),Weather_Condition,Amenity,Bump,Crossing,Give_Way,Junction,No_
Exit,Railway,Roundabout,Station,Stop,Traffic_Calming,Traffic_Signal,Turning_Loop,Sunrise_Sunset,Civil_Twilight,Nautical_
Twilight,Astronomical_Twilight
A-1 Source2 3 2016-02-08 05:46:00 2016-02-08 11:00:00 39.865147 -84.058723 -0.01 Right lane blocked due to accident on

```

6.5 Upload Dataset to HDFS

The US Accidents CSV dataset was uploaded from the local dataset folder into /usaccidents/input using `hdfs dfs -put` or `hdfs dfs -copyFromLocal`. After upload, the dataset became available to Hadoop jobs through the HDFS path rather than only through the local Windows filesystem.

```

C:\Windows\system32>hdfs dfs -chmod 777 /usaccidents
C:\Windows\system32>hdfs dfs -ls /usaccidents
Found 1 items
drwxr-xr-x - Tech Source supergroup 0 2026-09-06 23:28 /usaccidents/input

```

6.6 Verify Dataset in HDFS

The uploaded file was verified using HDFS listing and file inspection commands. This verification step ensures that the input exists at the expected path before running the analytical jobs. The HDFS input directory was reused by both MapReduce and Pig analyses.

```

C:\Windows\system32>hdfs dfs -ls /usaccidents
Found 1 items
drwxr-xr-x - Tech Source supergroup 0 2026-09-06 23:28 /usaccidents/input

```

7.MapReduce Analysis

Five Java MapReduce programs were prepared for the project. Each program follows the Mapper/Reducer processing model and is packaged into a JAR before execution. The jobs read the common HDFS input and write their results to separate HDFS output directories. The Java source files include StateCount.java, SeverityCount.java, CityCount.java, WeatherCount.java, and TrafficSignalCount.java.

The Hadoop compilation classpath included the common, HDFS, MapReduce, and YARN libraries. After compilation, the class files were packaged into JAR files such as StateCount.jar, SeverityCount.jar, CityCount.jar, WeatherCount.jar, and TrafficSignalCount.jar. The use of separate JARs keeps each analytical task simple and independently executable

1. State-wise Accident Count
2. Severity-wise Accident Count
3. City-wise Accident Count
4. Weather Condition Analysis
5. Traffic Signal-wise Accident Count

BROWSE DIRECTORY

/usaccidents

Go!

Show

25

entries

Search:

☐	Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name	
☐	drwxr-xr-x	Tech Source	supergroup	0 B	Sep 06 23:28	0	0 B	input	
☐	drwxr-xr-x	TechSource	supergroup	0 B	Sep 07 11:15	0	0 B	output_city	
☐	drwxr-xr-x	TechSource	supergroup	0 B	Sep 07 11:04	0	0 B	output_severity	
☐	drwxr-xr-x	TechSource	supergroup	0 B	Sep 07 11:38	0	0 B	output_signal	
☐	drwxr-xr-x	TechSource	supergroup	0 B	Sep 06 23:54	0	0 B	output_state	
☐	drwxr-xr-x	TechSource	supergroup	0 B	Sep 07 11:31	0	0 B	output_weather	

7.1 MapReduce Task 1: State-wise Accident Count

The StateCount program reads the State field and emits the state as the key with a count of one. The Reducer sums the values for each state. The output is written to /usaccidents/output_state. This analysis provides a state-level view of accident frequency.

```
C:\Windows\system32>hdfs dfs -ls /usaccidents/output_state
Found 2 items
-rw-r--r-- 1 TechSource supergroup          0 2026-09-06 23:54 /usaccidents/output_state/_SUCCESS
-rw-r--r-- 1 TechSource supergroup    405 2026-09-06 23:54 /usaccidents/output_state/part-r-00000

C:\Windows\system32>hdfs dfs -cat /usaccidents/output_state/part-r-00000
AL      33087
AR       984
AZ     22921
CA    336555
CO     13343
CT     11275
DC      1873
DE      2718
FL    142758
GA     45434
IA      5582
ID        70
IL     60311
IN      8304
KS      2244
```

7.2 MapReduce Task 2: Severity-wise Accident Count

The SeverityCount program extracts the Severity field and counts how many records belong to each severity category. This gives a compact distribution of accident seriousness across the dataset.

```
C:\BigDataProject\src>notepad SeverityCount.java

C:\BigDataProject\src>javac -cp "C:\hadoop\share\hadoop\common\*;C:\hadoop\share\hadoop\common\lib\*;C:\hadoop\share\hadoop\hdfs\*;C:\hadoop\share\hadoop\mapreduce\*;C:\hadoop\share\hadoop\yarn\*;C:\hadoop\share\hadoop\yarn\lib\*" -d ..\classes SeverityCount.java

C:\BigDataProject\src>dir ..\classes\SeverityCount*
Volume in drive C is Windows10
Volume Serial Number is 963B-8F97

Directory of C:\BigDataProject\classes

09/07/2026  11:00 AM                1,843 SeverityCount$SeverityMapper.class
09/07/2026  11:00 AM                1,755 SeverityCount$SeverityReducer.class
09/07/2026  11:00 AM                1,474 SeverityCount.class
               3 File(s)                5,072 bytes
               0 Dir(s) 42,453,887,184 bytes free

C:\BigDataProject\src>jar -cvf ..\SeverityCount.jar -C ..\classes .
added manifest
adding: SeverityCount$SeverityMapper.class(in = 1843) (out= 791)(deflated 57%)
adding: SeverityCount$SeverityReducer.class(in = 1755) (out= 742)(deflated 57%)
adding: SeverityCount.class(in = 1474) (out= 788)(deflated 46%)
adding: StateCount$StateMapper.class(in = 1917) (out= 845)(deflated 55%)
adding: StateCount$StateReducer.class(in = 1748) (out= 737)(deflated 57%)
adding: StateCount.class(in = 1627) (out= 902)(deflated 44%)
```

7.3 MapReduce Task 3: City-wise Accident Count

The CityCount program extracts City and counts accident records for each city. Because city names occur across many records, the Reducer aggregates identical city keys into total counts.

```

C:\BigDataProject\src>javac -cp "C:\hadoop\share\hadoop\common\*;C:\hadoop\share\hadoop\common\lib\*;C:\hadoop\share\hadoop\hdfs\*;C:\hadoop\share\hadoop\mapreduce\*;C:\hadoop\share\hadoop\yarn\*;C:\hadoop\share\hadoop\yarn\lib\*" -d ..\classes CityCount.java
javac: file not found: CityCount.java
Usage: javac <options> <source files>
Use -help for a list of possible options

C:\BigDataProject\src>dir
Volume in drive C is Windows10
Volume Serial Number is 963B-8F97

Directory of C:\BigDataProject\src

09/07/2026  11:10 AM    <DIR>          .
09/07/2026  11:01 AM    <DIR>          ..
09/07/2026  11:10 AM                2,532 CityCount.txt
09/07/2026  11:00 AM                2,388 SeverityCount.java
09/06/2026  11:38 PM                2,826 StateCount.java
               3 File(s)              7,746 bytes
               2 Dir(s)  42,468,601,856 bytes free

C:\BigDataProject\src>ren CityCount.txt CityCount.java

C:\BigDataProject\src>javac -cp "C:\hadoop\share\hadoop\common\*;C:\hadoop\share\hadoop\common\lib\*;C:\hadoop\share\hadoop\hdfs\*;C:\hadoop\share\hadoop\mapreduce\*;C:\hadoop\share\hadoop\yarn\*;C:\hadoop\share\hadoop\yarn\lib\*" -d ..\classes CityCount.java

```

7.4 MapReduce Task 4: Weather Condition Count

The WeatherCount program extracts Weather_Condition and counts records for each non-empty weather category. This helps summarize the environmental conditions appearing in the accident data.

```

C:\BigDataProject\src>jar -cvf ..\WeatherCount.jar -C ..\classes WeatherCount.class -C ..\classes WeatherCount$WeatherMapper.class -C ..\classes WeatherCount$WeatherReducer.class
added manifest
adding: WeatherCount.class(in = 1475) (out= 796)(deflated 46%)
adding: WeatherCount$WeatherMapper.class(in = 1885) (out= 835)(deflated 55%)
adding: WeatherCount$WeatherReducer.class(in = 1750) (out= 739)(deflated 57%)

C:\BigDataProject\src>hadoop jar ..\WeatherCount.jar WeatherCount /usaccidents/input /usaccidents/output_weather
2026-09-07 11:29:27,803 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-07 11:29:28,842 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-07 11:29:28,863 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Tech Source/.staging/job_1788716948664_0007
2026-09-07 11:29:29,175 INFO input.FileInputFormat: Total input files to process : 1
2026-09-07 11:29:29,266 INFO mapreduce.JobSubmitter: number of splits:5
2026-09-07 11:29:29,450 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1788716948664_0007
2026-09-07 11:29:29,453 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-07 11:29:29,730 INFO conf.Configuration: resource-types.xml not found
2026-09-07 11:29:29,731 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-07 11:29:29,841 INFO impl.YarnClientImpl: Submitted application application_1788716948664_0007
2026-09-07 11:29:29,886 INFO mapreduce.Job: The url to track the job: http://NASRIN:8088/proxy/application_1788716948664_0007/
2026-09-07 11:29:29,887 INFO mapreduce.Job: Running job: job_1788716948664_0007
2026-09-07 11:29:34,973 INFO mapreduce.Job: Job job_1788716948664_0007 running in uber mode : false

```

7.5 MapReduce Task 5: Traffic Signal Count

The TrafficSignalCount program uses the Traffic_Signal field and counts available category values. The program skips the CSV header and ignores empty values so that the aggregation

represents actual records rather than the column name.

```
C:\BigDataProject\src>notepad TrafficSignalCount.java

C:\BigDataProject\src>javac -cp "C:\hadoop\share\hadoop\common\*;C:\hadoop\share\hadoop\common\lib\*;C:\hadoop\share\hadoop\hdfs\*;C:\hadoop\share\hadoop\mapreduce\*;C:\hadoop\share\hadoop\yarn\*;C:\hadoop\share\hadoop\yarn\lib\*" -d ../classes TrafficSignalCount.java

C:\BigDataProject\src>jar -cvf ../TrafficSignalCount.jar -C ../classes TrafficSignalCount.class -C ../classes TrafficSignalCount$TrafficSignalMapper.class -C ../classes TrafficSignalCount$TrafficSignalReducer.class
added manifest
adding: TrafficSignalCount.class(in = 1520) (out= 796)(deflated 47%)
adding: TrafficSignalCount$TrafficSignalMapper.class(in = 1914) (out= 831)(deflated 56%)
adding: TrafficSignalCount$TrafficSignalReducer.class(in = 1780) (out= 745)(deflated 58%)

C:\BigDataProject\src>hadoop jar ../TrafficSignalCount.jar TrafficSignalCount /usaccidents/input /usaccidents/output_signal
2026-09-07 11:36:15,983 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-07 11:36:16,945 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-07 11:36:16,966 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging-1788716948664_0009
2026-09-07 11:36:17,236 INFO input.FileInputFormat: Total input files to process : 1
2026-09-07 11:36:17,328 INFO mapreduce.JobSubmitter: number of splits:5
```

Verified MapReduce Example: State-wise Output

The verified StateCount output contains the following selected state totals. These values are included as actual project output rather than estimated figures.

State	Accident Count
CA	336,555
TX	164,364
FL	142,758
SC	83,286

PA	64,569
NY	62,698
IL	60,311
NC	51,521
GA	45,434
MI	46,102

The complete verified output also includes all other states present in the dataset. California has the highest count among the listed results, followed by Texas and Florida, indicating a strong concentration of recorded accidents in these states.

8. Apache Pig Analysis

Apache Pig was used to perform ten additional analyses on the US Accidents data. Pig Latin provides a concise way to load the CSV with a defined schema, filter records, group records by one or more fields, calculate aggregates, and store results back to HDFS. Each task uses its own HDFS output path so that the results remain isolated and can be verified independently.

Ten pig analyses were performed:

1. Severe Accident Filtering
2. State-wise Accident Count
3. Severity-wise Accident Count

4. Weather Condition Analysis
5. City-wise Accident Count
6. State-wise Average Severity
7. Day and Night Accident Analysis
8. Severity-wise Average Distance
9. Timezone-wise Accident Count
10. Weather and Severity Analysis

Show 25 entries Search:

☐	Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name	
☐	drwxr-xr-x	TechSource	supergroup	0 B	Sep 08 00:04	0	0 B	task10_weather_severity	
☐	drwxr-xr-x	TechSource	supergroup	0 B	Sep 07 17:13	0	0 B	task1_severe	
☐	drwxr-xr-x	TechSource	supergroup	0 B	Sep 07 18:32	0	0 B	task2_state_count	
☐	drwxr-xr-x	TechSource	supergroup	0 B	Sep 07 19:05	0	0 B	task3_severity_count	
☐	drwxr-xr-x	TechSource	supergroup	0 B	Sep 07 19:40	0	0 B	task4_weather_count	
☐	drwxr-xr-x	TechSource	supergroup	0 B	Sep 07 20:09	0	0 B	task5_city_count	
☐	drwxr-xr-x	TechSource	supergroup	0 B	Sep 07 21:41	0	0 B	task6_state_avg_severity	
☐	drwxr-xr-x	TechSource	supergroup	0 B	Sep 07 22:21	0	0 B	task7_day_night	
☐	drwxr-xr-x	TechSource	supergroup	0 B	Sep 07 22:52	0	0 B	task8_severity_avg_distance	
☐	drwxr-xr-x	TechSource	supergroup	0 B	Sep 07 23:29	0	0 B	task9_timezone_count	

8.1 Pig Task 1: Severe Accidents

Filters records where Severity equals 4 and stores the resulting severe-accident records in /usaccidents/pig/task1_severe.

```
C:\BigDataProject\src>set HADOOP_USER_NAME=TechSource
C:\BigDataProject\src>echo %HADOOP_USER_NAME%
TechSource
C:\BigDataProject\src>hdfs dfs -ls -h /usaccidents/input/US_Accidents_1.5M.csv
-rw-r--r-- 1 Tech Source supergroup 540.7 M 2026-09-06 23:28 /usaccidents/input/US_Accidents_1.5M.csv
C:\BigDataProject\src>pig PigTask1.pig
2026-09-07 17:12:12,732 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-07 17:12:12,735 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-07 17:12:12,735 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-07 17:12:13,138 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-09-07 17:12:13,138 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788779533127.1
og
2026-09-07 17:12:13,825 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\Tech Source/.pigbootu
p not found
2026-09-07 17:12:13,924 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated
. Instead, use mapreduce.jobtracker.address
2026-09-07 17:12:13,925 [main] INFO org.apache.pig.backend.hadoop.executionengine.MExecutionEngine - Connecting to had
op file system at: hdfs://localhost:9000
2026-09-07 17:12:14,639 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-PigTask1.pig-0aa7bbd2
-e2b5-4a8a-9004-eae15e5e1c05
2026-09-07 17:12:14,639 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set
to false
2026-09-07 17:12:15,604 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separato
r is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-07 17:12:15,633 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: FILTER
```

Output for Severe Accidents

Block ID: 1073742199
Block Pool ID: BP-79721839-192.168.0.183-1787044306402
Generation Stamp: 1375
Size: 72525
Availability:
• NASRIN

File contents
CA,336555
TX,164364
FL,142758
SC,83286
PA,64569
NY,62698
IL,80311
NC,51521

8.2 Pig Task 2: State-wise Count

Groups the dataset by State and calculates the number of accidents in each state. Output path: /usaccidents/pig/task2_state_count.

```
C:\BigDataProject\src>pig PigTask2.pig
2026-09-07 18:20:10,444 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-07 18:20:10,447 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-07 18:20:10,447 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-07 18:20:10,851 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-09-07 18:20:10,851 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788783610840.10g
2026-09-07 18:20:11,470 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\Tech Source/.pigbootup not found
2026-09-07 18:20:11,545 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated . Instead, use mapreduce.jobtracker.address
2026-09-07 18:20:11,545 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-07 18:20:12,181 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-PigTask2.pig-10ec91d4-d4ad-4426-b3a1-bff4930867f8
2026-09-07 18:20:12,182 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-07 18:20:13,352 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-07 18:20:13,396 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,ORDER_BY,FILTER
2026-09-07 18:20:13,446 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-07 18:20:13,501 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten,
```

Output for State-wise Count

Block ID: 1073742238
Block Pool ID: BP-79721839-192.168.0.183-1787044306402
Generation Stamp: 1414
Size: 413
Availability:
• NASRIN

File contents
WY,3.52
MT,3.5
WV,3.136094674556213
MO,2.7159017220260826
CT,2.6878048780487807
KS,2.6818181818181817
ID,2.657142857142857
IN,2.647880539499037

8.3 Pig Task 3: Severity-wise Count

Groups records by Severity and counts the records in each severity category. Output path: /usaccidents/pig/task3_severity_count.

```
C:\BigDataProject\src>notepad PigTask3.pig

C:\BigDataProject\src>hdfs dfs -rm -r /usaccidents/pig/task3_severity_count
rm: '/usaccidents/pig/task3_severity_count': No such file or directory

C:\BigDataProject\src>set HADOOP_USER_NAME=TechSource

C:\BigDataProject\src>echo %HADOOP_USER_NAME%
TechSource

C:\BigDataProject\src>pig PigTask3.pig
2026-09-07 18:52:46,713 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-07 18:52:46,715 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-07 18:52:46,715 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-07 18:52:47,122 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-09-07 18:52:47,122 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788785567114.1
2026-09-07 18:52:47,768 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\Tech Source/.pigbootu
p not found
2026-09-07 18:52:47,840 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated
. Instead, use mapreduce.jobtracker.address
2026-09-07 18:52:47,840 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to had
op file system at: hdfs://localhost:9000
2026-09-07 18:52:48,492 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-PigTask3.pig-d02c2dbe
op file system at: hdfs://localhost:9000
```

Output for Severity-wise Count

Block ID: 1073742277

Block Pool ID: BP-79721839-192.168.0.183-1787044306402

Generation Stamp: 1453

Size: 33

Availability:

- NASRIN

File contents

r,False,False,False,False,False,False,False,False,False,False,False,False,Night,Night,Nig
ht,Night
A-708949,Source2,4,2022-02-15 05:07:45,2022-02-15
06:41:56,31.53224,-81.445053,,,15.229999542236328,Left hand shoulder blocked and left lane
blocked due to crash on I-95 Northbound between Exit 49 GA-251 and Exit 58 GA-57.,GA
Highway 57,Townsend,McIntosh,GA,31331,US,US/Eastern,KBQK,2022-02-15
05:15:00,34.0,34.0,88.0,30.42,10.0,CALM,0.0,0.0,Fair,False,False,False,False,False,False,
False,False,False,False,False,False,Night,Night,Night,Night

8.4 Pig Task 4: Weather Condition Count

Groups records by Weather_Condition and counts the frequency of each weather condition. Output path: /usaccidents/pig/task4_weather_count.

```

C:\BigDataProject\src>set HADOOP_USER_NAME=TechSource

C:\BigDataProject\src>echo %HADOOP_USER_NAME%
TechSource

C:\BigDataProject\src>pig PigTask4.pig
2026-09-07 19:27:57,693 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-07 19:27:57,694 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-07 19:27:57,694 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-07 19:27:58,107 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-09-07 19:27:58,107 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788787678102.107
2026-09-07 19:27:58,715 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\Tech Source/.pigbootup not found
2026-09-07 19:27:58,791 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-07 19:27:58,791 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-07 19:27:59,432 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-PigTask4.pig-00013dce-d612-4f3f-a9e6-a43fdb82fc29
2026-09-07 19:27:59,432 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-07 19:28:00,480 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-07 19:28:00,516 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY, ORDER_BY, FILTER
2026-09-07 19:28:00,553 [main] INFO org.apache.pig.data.SchemaToolBackend - You can experimentally use set set ... will

```

Output for Weather Condition Count

Block ID: 1073742316

Block Pool ID: BP-79721839-192.168.0.183-1787044306402

Generation Stamp: 1492

Size: 2298

Availability:

- NASRIN

File contents

```

Houston,53364
Dallas,37963
Los Angeles,33874
Austin,27416
Charlotte,26184
Atlanta,19132
Miami,15465
Orlando,14969

```

8.5 Pig Task 5: City-wise Count

Groups records by City and counts accidents for each city. Output path: /usaccidents/pig/task5_city_count.

```

C:\BigDataProject\src>set HADOOP_USER_NAME=TechSource

C:\BigDataProject\src>echo %HADOOP_USER_NAME%
TechSource

C:\BigDataProject\src>pig PigTask5.pig
2026-09-07 19:58:09,462 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-07 19:58:09,462 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-07 19:58:09,464 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-07 19:58:09,666 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-09-07 19:58:09,666 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788789489661.107
2026-09-07 19:58:10,018 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\Tech Source/.pigbootup not found
2026-09-07 19:58:10,062 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated

```

Output for City-wise Count

Block ID: 1073742355
Block Pool ID: BP-79721839-192.168.0.183-1787044306402
Generation Stamp: 1531
Size: 126152
Availability:
• NASRIN

File contents
CA,336555
TX,164364
FL,142758
SC,83286
PA,64569
NY,62698
IL,60311
NC,51521

8.6 Pig Task 6: Average Severity by State

Groups records by State and calculates AVG(Severity), providing an average severity value for each state. Output path: /usaccidents/pig/task6_state_avg_severity.

```
C:\BigDataProject\src>set HADOOP_USER_NAME=TechSource
C:\BigDataProject\src>echo %HADOOP_USER_NAME%
TechSource
C:\BigDataProject\src>pig PigTask6.pig
2026-09-07 21:29:03,077 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-07 21:29:03,078 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-07 21:29:03,078 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-07 21:29:03,277 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-09-07 21:29:03,277 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788794943272.1
og
2026-09-07 21:29:03,636 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\Tech Source/.pigbootu
p not found
2026-09-07 21:29:03,678 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated
. Instead, use mapreduce.jobtracker.address
2026-09-07 21:29:03,678 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to had
op file system at: hdfs://localhost:9000
2026-09-07 21:29:04,049 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-PigTask6.pig-03accf94 ▼
```

Output for Average Severity by State

Block ID: 1073742394
Block Pool ID: BP-79721839-192.168.0.183-1787044306402
Generation Stamp: 1570
Size: 998
Availability:
• NASRIN

File contents
1,0.05192727275013149
2,0.09673681117568761
3,0.3528305848709385
4,4.801327574559954

8.7 Pig Task 7: Day/Night Accident Count

Uses Sunrise_Sunset to group records into day/night categories and counts accidents for each category. Output path: /usaccidents/pig/task7_day_night.

```
C:\BigDataProject\src>pig PigTask7.pig
2026-09-07 22:10:09,646 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-07 22:10:09,647 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-07 22:10:09,647 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-07 22:10:09,841 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-09-07 22:10:09,842 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_17887974089837.1
2026-09-07 22:10:10,161 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\Tech Source/.pigbootup
2026-09-07 22:10:10,200 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated
. Instead, use mapreduce.jobtracker.address
2026-09-07 22:10:10,200 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop
file system at: hdfs://localhost:9000
2026-09-07 22:10:10,573 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-PigTask7.pig-eb551a2c
```

Output for Day/Night Accident Count

Block ID: 1073742433
Block Pool ID: BP-79721839-192.168.0.183-1787044306402
Generation Stamp: 1609
Size: 42
Availability:
• NASRIN

File contents
Fair,466102
Clear,208905
Mostly Cloudy,201302
Cloudy,147317
Partly Cloudy,136487
Overcast,80265
Light Rain,65383
Scattered Clouds,47595

8.8 Pig Task 8: Average Distance by Severity

Groups records by Severity and calculates average Distance for each severity level. Output path: /usaccidents/pig/task8_severity_avg_distance.

```

C:\BigDataProject\src>pig PigTask8.pig
2026-09-07 22:40:51,347 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-07 22:40:51,349 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-07 22:40:51,349 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-07 22:40:51,546 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017:41:58
2026-09-07 22:40:51,546 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_17887992510g
2026-09-07 22:40:51,867 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\Tech Source/.pigbootu
p not found
2026-09-07 22:40:51,905 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is depre
. Instead, use mapreduce.jobtracker.address
2026-09-07 22:40:51,906 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to
op file system at: hdfs://localhost:9000
2026-09-07 22:40:52,280 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-PigTask8.pig-3d5
-b136-42fe-92f6-65fb9a74717a
2026-09-07 22:40:52,281 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled
to false
2026-09-07 22:40:52,816 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.sepa
r is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-07 22:40:52,838 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: G
BY,ORDER_BY,FILTER
2026-09-07 22:40:52,861 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set...
not generate code.

```

Output for Average Distance by Severity

Block ID: 1073742472
Block Pool ID: BP-79721839-192.168.0.183-1787044306402
Generation Stamp: 1648
Size: 85
Availability:
• NASRIN

File contents
<pre> ver Spring, Montgomery, MD, 20910, US, US/Eastern, KCGS, 2021-04-25 07:15:00,49.0,46.0,90.0,29.65,5.0,N,6.0,0.0,Cloudy,False,False,True,False,False,Fal se,False,False,False,False,False,Day,Day,Day,Day A-1069896,Source2,4,2021-04-25 17:20:22,2021-04-25 18:34:31,39.420651,-77.484131,,0.83999999737739563,One lane blocked due to accident on I-70 Eastbound from Exit 48 US-40 to Exit 49 US-40-ALT, I-70 E, Frederick, Frederick, MD, 21703, US, US/Eastern, KFDK, 2021-04-25 17:47:00,64.0,64.0,45.0,29.53,10.0,NW,15.0,0.0,Mostly </pre>

8.9 Pig Task 9: Timezone-wise Count

Groups records by Timezone and counts accidents in each timezone category. Output path: /usaccidents/pig/task9_timezone_count.

```

C:\BigDataProject\src>pig PigTask9.pig
2026-09-07 23:16:19,253 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-07 23:16:19,254 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-07 23:16:19,255 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-07 23:16:19,710 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-09-07 23:16:19,710 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788801379702.10g
2026-09-07 23:16:20,305 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\Tech Source/.pigbootu
p not found
2026-09-07 23:16:20,370 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated
. Instead, use mapreduce.jobtracker.address
2026-09-07 23:16:20,370 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to had
op file system at: hdfs://localhost:9000
2026-09-07 23:16:20,983 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-PigTask9.pig-a5be0b1f
-ea85-44b2-aad9-79d3e45e74bc
2026-09-07 23:16:20,983 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set
to false
2026-09-07 23:16:21,897 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separa
r is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-07 23:16:21,930 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_
BY,ORDER_BY,FILTER
2026-09-07 23:16:21,968 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will
not generate code.
2026-09-07 23:16:22,016 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[Add
ForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeF
ilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten,
PushUpFilter, SplitFilter, SplitTypeCastInserter]}

```

Output for Timezone-wise Count

Block ID: 1073742511
Block Pool ID: BP-79721839-192.168.0.183-1787044306402
Generation Stamp: 1687
Size: 83
Availability:
• NASRIN

File contents
Day,1092573
Night,405782
Sunrise_Sunset,1

8.10 Pig Task 10: Weather Condition and Severity

Groups records by the combination of Weather_Condition and Severity and counts each combination. Output path: /usaccidents/pig/task10_weather_severity.

```
C:\BigDataProject\src>pig PigTask10.pig
2026-09-07 23:51:30,805 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-07 23:51:30,807 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-07 23:51:30,808 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-07 23:51:31,384 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-09-07 23:51:31,384 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788803491371.1
2026-09-07 23:51:32,265 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\Tech Source/.pigbootu
p not found
2026-09-07 23:51:32,368 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated
. Instead, use mapreduce.jobtracker.address
2026-09-07 23:51:32,368 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to had
op file system at: hdfs://localhost:9000
2026-09-07 23:51:33,515 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-PigTask10.pig-9d85f3b
6-e5ff-40ed-a729-38e1164c38a7
2026-09-07 23:51:33,515 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set
to false
2026-09-07 23:51:35,193 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator
is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-07 23:51:35,236 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_
BY,ORDER_BY,FILTER
2026-09-07 23:51:35,293 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will
not generate code.
2026-09-07 23:51:35,364 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[Add
ForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeF
```

Output for Weather Condition and Severity

Block ID: 1073742550
Block Pool ID: BP-79721839-192.168.0.183-1787044306402
Generation Stamp: 1726
Size: 6143
Availability:
• NASRIN

File contents
Day,1092573
Night,405782
Sunrise_Sunset,1

9.Results and Discussion

The project successfully produced multiple MapReduce and Pig outputs from the same HDFS dataset. These analyses examine accidents from different dimensions, including location, severity, weather, distance, time, timezone, and traffic infrastructure. Separate HDFS output directories allow each result to be checked independently.

The verified state-wise MapReduce result shows significant differences among states. **California recorded 336,555 accidents, Texas 164,364, and Florida 142,758**, while South Carolina, Pennsylvania, and New York also showed high counts. This demonstrates that accident records are not evenly distributed geographically.

The severity, city, and weather analyses provide additional information about accident patterns. Pig tasks such as **average severity, average distance, day/night, timezone, and weather-severity analysis** provide further descriptive insights into the dataset.

Overall, **MapReduce and Pig provide complementary approaches**. MapReduce offers detailed programming control, while Pig simplifies common filtering, grouping, counting, and aggregation tasks. The project therefore demonstrates practical Big Data analysis using **HDFS, MapReduce, and Apache Pig** on a real-world dataset.

10.Limitations and Future Scope

- The analysis is primarily descriptive and does not establish causal relationships between weather, road features, and accidents.
- CSV parsing with simple comma-delimited storage can be sensitive to commas inside text fields; a production pipeline would benefit from robust CSV parsing.
- Some fields contain missing values, so filtering and aggregation need careful handling.
- City-wise and weather-wise grouping can produce many unique categories, which may make direct manual inspection difficult.
- The project can be extended with Apache Hive for SQL-style queries and Apache Spark for faster in-memory distributed analytics.
- Future work could include visualization dashboards, time-series trend analysis, geospatial clustering, anomaly detection, and predictive modeling.
- Data-quality validation could be expanded to detect duplicate records, invalid coordinates, impossible durations, and inconsistent timestamps.

11.Conclusion

This project successfully demonstrates an end-to-end Big Data processing workflow using the US Accidents dataset and the Hadoop ecosystem. The dataset was prepared, uploaded to HDFS, and used as the common source for distributed analysis. This workflow shows how large-scale raw data can be stored and processed using Hadoop technologies.

The MapReduce component includes five Java applications for analyzing accidents by state, severity, city, weather condition, and traffic signals. The verified results show meaningful accident distributions, including 336,555 records in California, 164,364 in Texas, and 142,758 in Florida.

The Apache Pig component includes ten tasks covering filtering, grouping, counting, averaging, and combined analysis. These tasks provide additional insights into accident severity, location, weather, distance, time, and timezone.

Overall, the project demonstrates the complementary roles of HDFS, MapReduce, and Apache Pig in Big Data analytics. It connects theoretical concepts with practical analysis of a real-world dataset and provides a foundation for future work using Hive, Spark, visualization, and machine learning.

12.References

1. **Hadoop Documentation** — Apache Hadoop, <https://hadoop.apache.org/>
2. **Apache Pig Documentation** — Apache Pig, <https://pig.apache.org/>
3. US Accidents dataset and accompanying dataset documentation used for this project.
4. **Project source files:** StateCount.java, SeverityCount.java, CityCount.java, WeatherCount.java, TrafficSignalCount.java.
5. **Project Pig scripts:** PigTask1.pig through PigTask10.pig.
6. <https://github.com/SiddhiPrabhu1995/YouTube-Data-Analysis-using-Hadoop/tree/master/PigAnalysis/scripts>